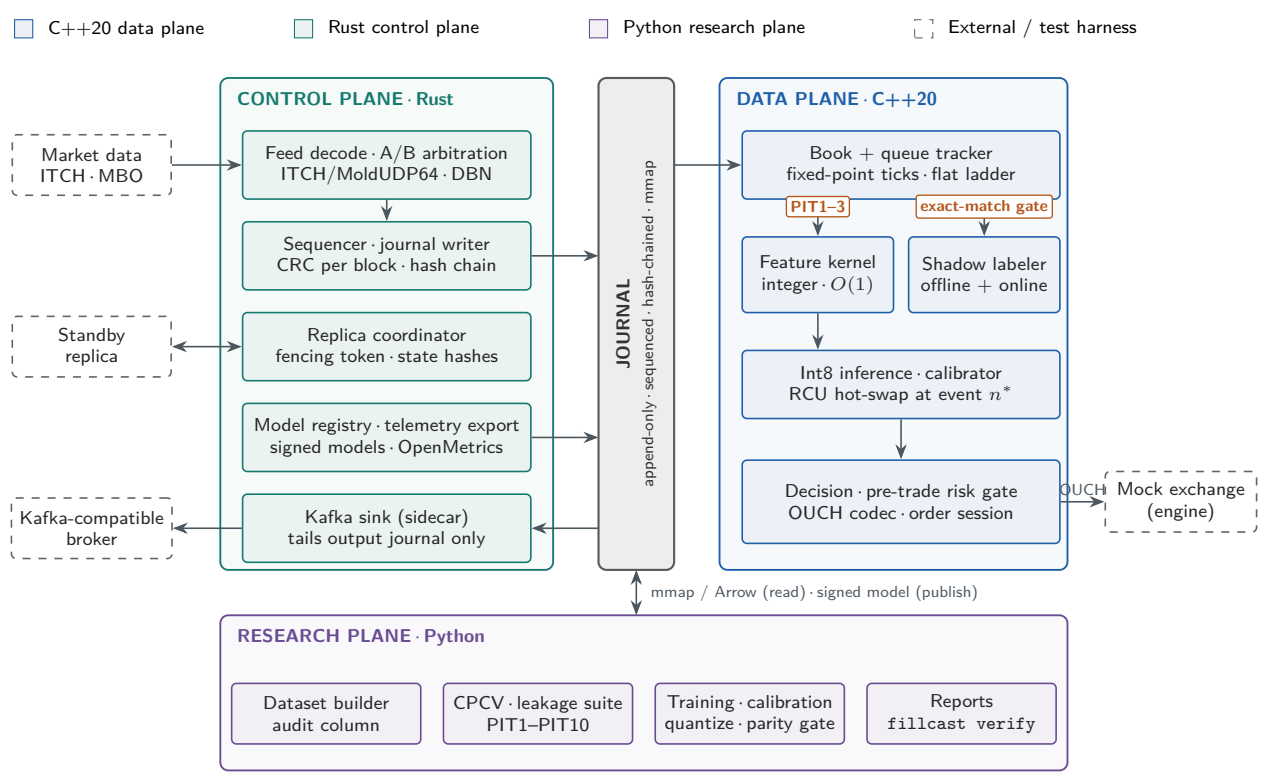
C++20 data plane
Rust control plane
Python research plane
External / test harness
CONTROL PLANE · Rust
Market data
ITCH · MBO
Feed decode · A/B arbitration
ITCH/MoldUDP64 · DBN
Sequencer · journal writer
CRC per block · hash chain
Standby
replica
Replica coordinator
fencing token · state hashes
Model registry · telemetry export
signed models · OpenMetrics
Kafka-compatible
broker
Kafka sink (sidecar)
tails output journal only
JOURNAL
append-only · sequenced · hash-chained · mmap
DATA PLANE · C++20
Book + queue tracker
fixed-point ticks · flat ladder
PIT1–3
exact-match gate
Feature kernel
integer · $O(1)$
Shadow labeler
offline + online
Int8 inference · calibrator
RCU hot-swap at event n^*
Decision · pre-trade risk gate
OUCH codec · order session
OUCH
Mock exchange
(engine)
mmap / Arrow (read) · signed model (publish)
RESEARCH PLANE · Python
Dataset builder
audit column
CPCV · leakage suite
PIT1–PIT10
Training · calibration
quantize · parity gate
Reports
fillcast verify